# Feature Detection for SLAM

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# 1 Notation, Conventions, and Basic Models

## 1.1 Discrete image model and coordinates

**Definition 1.1** (Image as a discrete signal)**.** An image is a function

$$I : \Omega \subset \mathbb{Z}^2 \to \mathbb{R}$$

Mapping pixel coordinates $x = (u, v) \in \Omega$ to an intensity value $I(x)$. In practice, $I$ may be single-channel (grayscale) or a vector-valued map (colour); in these notes we will work with a scalar field.

*Remark* 1.2. Although $I$ is discrete, most derivations become cleaner by passing to a smooth interpolation (e.g. bilinear) and treating $I$ locally as a differentiable function on $\mathbb{R}^2$. We will state explicitly when we make this transition.

## 1.2 Local patches and windows

**Definition 1.3** (Patch / window)**.** Let $x \in \Omega$ and let $\mathcal{W} \subset \mathbb{Z}^2$ be a finite, symmetric index set (e.g. a square).

The patch of $I$ centred at $x$ over $\mathcal{W}$ is the collection

$$\{I(x + r) : r \in \mathcal{W}\}$$

We will write sums over a window as: $\sum_{r \in \mathcal{W}}$

## 1.3 Interpolation and gradients

**Assumption 1.4** (Differentiable local model)**.** When deriving gradient-based identities, we assume $I$ is extended to a differentiable function $\tilde{I} : \mathbb{R}^2 \to \mathbb{R}$ via an interpolation operator (e.g. bilinear or bicubic), and write $I$ for $\tilde{I}$ by assumption.

**Definition 1.5** (Spatial gradient)**.** The (continuous) spatial gradient of $I$ at $x \in \mathbb{R}^2$ is

$$\boldsymbol{\nabla} I(x) := \begin{bmatrix} \partial_u I(x) \\ \partial_v I(x) \end{bmatrix} \in \mathbb{R}^2$$

In discrete implementations, $\partial_u, \partial_v$ are approximated by finite differences or derivative-of-Gaussian filters.

## 1.4 Photometric assumptions for tracking

**Assumption 1.6** (Brightness constancy)**.** Let $x_t$ denote a physical scene point projected at time $t$ to image coordinate $x_t \in \mathbb{R}^2$. Over a short time interval, the observed intensity is approximately constant:

$$I_t(x_t) \approx I_{t+\Delta t}(x_{t+\Delta t})$$

*Remark* 1.7. Brightness constancy is violated by exposure changes, specularities, motion blur, rolling shutter, for example. For MonoSLAM-style front-ends, it remains an effective local model when combined with robust feature selection and outlier rejection.

## 1.5 Small-displacement linearisation

**Proposition 1.8** (First-order Taylor approximation under translation)**.** *Let $I : \mathbb{R}^2 \to \mathbb{R}$ be differentiable. For a fixed point $x \in \mathbb{R}^2$ and a small displacement $\delta \in \mathbb{R}^2$ we have*

$$I(x + \delta) \;=\; I(x) + \boldsymbol{\nabla} I(x)^{\mathsf{T}} \delta \;+\; O(\|\delta\|^2)$$

*Proof.* This is the standard multivariate Taylor expansion of $I$ about $x$:

$$I(x+\delta) = I(x) + DI(x)[\delta] + \frac{1}{2}D^2I(x)[\delta,\delta] + \cdots$$

Where the first differential satisfies $DI(x)[\delta] = \boldsymbol{\nabla} I(x)^{\mathsf{T}}\delta$ and the remainder is $O(\|\delta\|^2)$ as $\|\delta\| \to 0$ □

# 2 Image Formation and Pre-processing

## 2.1 A simple acquisition model

**Assumption 2.1** (Additive noise model)**.** We model an observed image $I$ as

$$I(x) = I^\star(x) + \eta(x)$$

Where $I^\star$ is an ideal noise-free image, and $\eta$ is random noise with $\mathbb{E}[\eta(x)] = 0$ and $\mathrm{Var}(\eta(x)) = \sigma^2$.

*Remark* 2.2. This is not a complete model of camera noise (i.e. image shot noise is signal-dependent, and read noise is not i.i.d. Gaussian), but it's a sufficient approximation to justify smoothing and to quantify variance reduction.

## 2.2 Smoothing as convolution

**Definition 2.3** (Discrete convolution with a kernel)**.** Let $k : \mathbb{Z}^2 \to \mathbb{R}$ be a finite-support kernel. The convolution $(k * I)$ is defined by

$$(k * I)(x) \coloneqq \sum_{r\in\mathbb{Z}^2} k(r)\, I(x-r)$$

Where the sum is effectively over the support of $k$.

**Definition 2.4** (Gaussian kernel)**.** For $\sigma > 0$, the (continuous) isotropic Gaussian is

$$G_\sigma(\xi) \coloneqq \frac{1}{2\pi\sigma^2}\exp\Big(-\frac{\|\xi\|^2}{2\sigma^2}\Big), \qquad \xi \in \mathbb{R}^2$$

Isotropic means rotation-invariant: $G_\sigma(\xi)$ depends only on the radius $\|\xi\|$, so its level sets are circles.

*Remark* 2.5. A discrete kernel is formed by sampling $G_\sigma$ on $\mathbb{Z}^2$ over a finite window (e.g. $\{-r,\dots,r\}^2$) and renormalising the sampled weights to sum to 1, so that convolution preserves average intensity.

**Proposition 2.6** (Gaussian separability)**.** *The 2D Gaussian factorises into a product of 1D Gaussians:*

$$G_\sigma(u,v) = g_\sigma(u)\, g_\sigma(v), \qquad g_\sigma(t) \coloneqq \frac{1}{\sqrt{2\pi}\sigma}\exp\Big(-\frac{t^2}{2\sigma^2}\Big)$$

*Proof.* By direct algebra,

$$G_\sigma(u,v) = \frac{1}{2\pi\sigma^2}\exp\Big(-\frac{u^2+v^2}{2\sigma^2}\Big) = \Big(\frac{1}{\sqrt{2\pi}\sigma}\exp\Big(-\frac{u^2}{2\sigma^2}\Big)\Big)\Big(\frac{1}{\sqrt{2\pi}\sigma}\exp\Big(-\frac{v^2}{2\sigma^2}\Big)\Big) = g_\sigma(u)\, g_\sigma(v)$$

□

*Remark* 2.7. For convolution, Fubini's theorem yields $(G_\sigma * I) = g_\sigma * (g_\sigma * I)$, where the inner convolution is along one axis and the outer along the other.

## 2.3 Variance reduction under smoothing

**Proposition 2.8** (Smoothing reduces noise variance)**.** *Assume $I = I^\star + \eta$ and consider a linear filter with kernel $k$ satisfying $\sum_r k(r) = 1$. If $\{\eta(x)\}$ are independent with $\mathbb{E}[\eta(x)] = 0$ and $\mathrm{Var}(\eta(x)) = \sigma^2$, then the filtered noise*

$$\eta_k(x) \coloneqq (k * \eta)(x) = \sum_r k(r)\, \eta(x - r)$$

*satisfies*

$$\mathbb{E}[\eta_k(x)] = 0, \qquad \mathrm{Var}(\eta_k(x)) = \sigma^2 \sum_r k(r)^2 \;\le\; \sigma^2$$

*Proof.* Linearity implies $\mathbb{E}[\eta_k(x)] = \sum_r k(r)\mathbb{E}[\eta(x - r)] = 0$

Independence gives

$$\mathrm{Var}(\eta_k(x)) = \sum_r k(r)^2\, \mathrm{Var}(\eta(x - r)) = \sigma^2 \sum_r k(r)^2$$

Since $\sum_r k(r) = 1$, the Jensen/Cauchy–Schwarz inequality implies $\sum_r k(r)^2 \le (\sum_r |k(r)|)^2 = 1$ when $k(r) \ge 0$ (as for Gaussian kernels), hence $\mathrm{Var}(\eta_k(x)) \le \sigma^2$ □

*Remark* 2.9. Variance reduction is only one part of the story: smoothing also attenuates high-frequency structure, which can degrade localisation. MonoSLAM front-ends therefore smooth mildly (often implicitly via derivative-of-Gaussian gradients) rather than aggressively.

## 2.4 Aliasing and blur-before-subsample

**Definition 2.10** (Downsampling by an integer factor)**.** Let $s \in \{2, 3, \dots\}$. The downsampled image $I^{\downarrow s}$ is defined on the grid $\Omega^{\downarrow s}$ by

$$I^{\downarrow s}(x) \coloneqq I(sx), \qquad x \in \Omega^{\downarrow s} \subset \mathbb{Z}^2$$

**Proposition 2.11** (Anti-aliasing (pre-blur))**.** *Downsampling by factor $s$ folds high-frequency content into low frequencies (aliasing). Applying a low-pass filter (e.g. Gaussian smoothing) with cutoff adapted to $s$ before downsampling reduces aliasing artefacts.*

*Proof.* A complete proof uses discrete-time Fourier analysis: downsampling corresponds to periodic replication and overlap in the frequency domain. When the original spectrum is not band-limited to the Nyquist region of the downsampled grid, these replications overlap, producing aliasing. Pre-filtering attenuates energy outside the admissible band, reducing overlap. □

*Remark* 2.12. High-frequency components are regions where intensity changes significantly. For example, edges, corners, and fine texture.

## 2.5 Gaussian pyramids

**Definition 2.13** (Gaussian pyramid)**.** Fix a smoothing kernel $G_\sigma$ and a downsampling factor $s = 2$. Define

$$I^{(0)} \coloneqq I, \qquad I^{(\ell+1)} \coloneqq \left(G_\sigma * I^{(\ell)}\right)^{\downarrow 2}, \qquad \ell = 0, 1, 2, \dots$$

The sequence $\{I^{(\ell)}\}$ is called a Gaussian pyramid.

*Remark* 2.14. For tracking (Lucas–Kanade / KLT), pyramids enable coarse-to-fine optimisation, effectively enlarging the basin of convergence for small-motion linearisations such as Proposition 1.8.

# 3 Feature Detection

## 3.1 Repeatability, localisation, and trackability

**Definition 3.1** (Repeatability (operational))**.** A detector is repeatable if, under admissible changes in viewpoint and photometry, it selects corresponding physical points across frames with high probability.

**Definition 3.2** (Localisation accuracy)**.** A feature is well-localised if its estimated position $\hat{x}$ has small uncertainty (e.g. subpixel) and small bias relative to the true image position.

**Definition 3.3** (Trackability)**.** A feature at location $x$ is trackable if small inter-frame motion can be estimated reliably from local image content (typically from a patch around $x$).

## 3.2 Lucas–Kanade optical flow as local least squares

**Assumption 3.4** (Brightness constancy on a window)**.** Let $W$ be a finite window of pixels around $x$. Assume that for a small displacement $u \in \mathbb{R}^2$,

$$I(p+u) \approx I(p), \qquad \forall p \in W$$

**Definition 3.5** (Lucas–Kanade local displacement estimate)**.** Under Assumption 3.4 and the first-order Taylor approximation (Proposition 1.8), define the residual at pixel $p \in W$ by

$$r_p(u) \coloneqq I(p+u) - I(p) \approx \boldsymbol{\nabla} I(p)^{\mathsf{T}} u$$

The Lucas–Kanade estimate is the least-squares minimiser

$$\hat{u} \in \arg\min_{u \in \mathbb{R}^2} \sum_{p \in W} \left(\boldsymbol{\nabla} I(p)^{\mathsf{T}} u\right)^2$$

**Proposition 3.6** (Normal equations and the structure tensor)**.** *The minimiser in Definition 3.5 satisfies the normal equations*

$$G\,u = 0, \qquad G \coloneqq \sum_{p \in W} \boldsymbol{\nabla} I(p)\, \boldsymbol{\nabla} I(p)^{\mathsf{T}} \in \mathbb{R}^{2\times 2}$$

*More generally, in the presence of a temporal term $I_t$ (optical flow),*

$$G\,u = b, \qquad b \coloneqq -\sum_{p \in W} \boldsymbol{\nabla} I(p)\, I_t(p)$$

## 3.3 Aperture problem and conditioning

**Proposition 3.7** (Edge ambiguity (aperture problem, local statement))**.** *Suppose that in a neighbourhood of $x$, the image intensity varies only along a single direction, i.e. $\boldsymbol{\nabla} I(x)$ is nonzero but nearly constant and rank-1 information dominates. Then, using only local brightness constancy over a small window, the displacement component orthogonal to $\boldsymbol{\nabla} I(x)$ is weakly constrained, leading to ambiguous motion estimates.*

*Proof.* Under brightness constancy and the first-order approximation (Proposition 1.8), the change in intensity for a small displacement $\delta$ is approximately $\boldsymbol{\nabla} I(x)^{\mathsf{T}} \delta$. If local gradients within a window span essentially a 1D subspace, then the data term depends primarily on the projection of $\delta$ onto that subspace, leaving the orthogonal component poorly observed. Formally, the normal equations in Lucas–Kanade involve a matrix of summed outer products of gradients; when these are nearly rank-1, the system is ill-conditioned and the solution is not stable to noise. □

### 3.4 The second-moment matrix (structure tensor) viewpoint

**Definition 3.8** (Second-moment matrix / structure tensor)**.** Given a window $\mathcal{W}$ and nonnegative weights $w(r)$, define the structure tensor

$$M(x) := \sum_{r \in \mathcal{W}} w(r)\, \boldsymbol{\nabla} I(x+r)\, \boldsymbol{\nabla} I(x+r)^{\mathsf{T}} \;\in\; \mathbb{R}^{2\times 2}$$

**Proposition 3.9** (Interpretation via eigenvalues)**.** *Let $\lambda_1(x) \geq \lambda_2(x) \geq 0$ denote the eigenvalues of $M(x)$. Then:*

- *If $\lambda_1 \approx \lambda_2 \approx 0$, the region is locally flat (little texture).*
- *If $\lambda_1 \gg \lambda_2 \approx 0$, the region resembles an edge (1D texture).*
- *If $\lambda_1$ and $\lambda_2$ are both large, the region is corner-like (2D texture), and local motion estimation is well-conditioned.*

*Proof.* Each term $\boldsymbol{\nabla} I, \boldsymbol{\nabla} I^{\mathsf{T}}$ is positive semidefinite, so $M(x)$ is positive semidefinite and has nonnegative eigenvalues. If all gradients are near zero within the window, then $M(x) \approx 0$ and both eigenvalues are small. If gradients are nonzero but all aligned with a single direction $g$, then $M(x) \approx c\, gg^{\mathsf{T}}$ for some $c > 0$, which is rank-1, hence one eigenvalue is large and the other near zero. If gradients span two independent directions with substantial energy, then $M(x)$ is full-rank and both eigenvalues are large. □

*Remark* 3.10 (SLAM relevance). In MonoSLAM-style EKF updates or bundle adjustment, poorly conditioned features (small $\lambda_2$) behave like "sliding" constraints and degrade robustness. Hence a practical policy is to favour points with large minimum eigenvalue (Shi–Tomasi criterion), subject to spatial distribution constraints - like corners.